Table 2. The table shows a comparison of emcee/AIES, emcee/DEMC and zeus in terms of the inverse efficiency (i.e. reciprocal of the number of independent samples per model evaluation or the autocorrelation time estimate times the average number of model evaluations per iteration per walker), the convergence cost (i.e. number of model evaluations until convergence) and the convergence fraction (i.e. fraction of converged chains for given maximum number of model evaluations).

	emcee/AIES	emcee/DEMC	**zeus**
Cosmological inference			
efficiency^{-1}	12140	6750	**1320**
convergence cost	24×10^5	22×10^5	$\mathbf{6.6 \times 10^5}$
convergence fraction	7/40	14/40	**36/40**
Exoplanet inference			
efficiency^{-1}	1386	338	**47**
convergence cost	36.0×10^2	17.1×10^2	$\mathbf{4.8 \times 10^2}$
convergence fraction	23/40	29/40	**38/40**

3.2.1 Cosmological inference

The particular inference problem that we consider here is that of the anisotropic BAO parameter inference using estimates of the galaxy power spectrum. The data we used come from the 12th data release (DR12) of the high-redshift North Galactic Cap (NGC) sample as observed by the Sloan Digital Sky Survey (SDSS) (Eisenstein et al. 2011) Baryon Oscillation Spectroscopic Survey (BOSS) (Dawson et al. 2013). Our analysis follows closely that of Beutler et al. (2017) with the difference that we chose not to fix any parameters and fit the hexadecapole multipole of the power spectrum as well as the monopole and quadrupole. Those choices were made solely to render the problem more challenging. Indeed the inclusion of the hexadecapole does not contribute any additional constraining power for the data that we used. However, such extended models will prove useful when analysing data from larger galaxy surveys such as DESI (DESI Collaboration et al. 2016). In terms of Bayesian inference, the problem has 22 free parameters. The results of our analysis are consistent with those of Beutler et al. (2017). We used weakly informative flat (uniform) priors for all parameters except for the two scaling parameters, $\alpha_{\parallel}$ and $\alpha_{\perp}$ for which we used normal (Gaussian) priors. We used 50 walkers in total.

In terms of efficiency, zeus generates at least 5 independent samples for each one generated by emcee/DEMC and at least 9 for each one generated by emcee/AIES factoring in the different computational cost of the methods. As for the convergence rate, zeus converges more than 3 times faster than either emcee variant. Finally, we found that zeus is less sensitive to the initialisation than either of the other two methods. In particular, out of the 40 tests conducted with different initialisation, zeus converged 36 times, emcee/DEMC 14 times and emcee/AIES 7 times prior to the predetermined maximum number of likelihood evaluations (i.e. 5×10^6 in this case). The aforementioned results are presented in detail in Table 2. The 1-D and 2-D marginal posterior distributions are shown in Figure 16 demonstrating the agreement between the three methods[4].

[4] No upper limit on the number of likelihood evaluations or iterations was used for this run and convergence was diagnosed using all the metrics that we introduced.

3.2.2 Exoplanet inference

Another common application of MCMC methods in astronomy is the problem of exoplanet parameter inference through modelling of Keplerian orbits and radial velocity time series data. In this section we demonstrate the performance of zeus using a two-planet model with 14 free parameters and real data from the K2-24 (EPIC-203771098) extrasolar system (Petigura et al. 2016) that is known to host two exoplanets. We used the popular Python package RadVel (Fulton et al. 2018) for the Keplerian modelling of the planetary orbits. The results of our analysis are consistent with published constraints for the aforementioned extrasolar system (Petigura et al. 2016). We used 30 walkers in total for sampling.

We performed the same suite of tests as in the cosmological inference case. In terms of efficiency, zeus generates more than 7 independent samples per each one generated by emcee/DEMC and more than 29 independent samples per each one generated by emcee/AIES. As for the convergence rate, zeus converges 7.5 times faster than emcee/AIES and 3.5 faster than emcee/DEMC on average. Finally, we found again that zeus is less sensitive to the specific initialisation of the walkers. In particular, out of the 40 tests conducted with different initialisation, zeus converged 38 times, emcee/DEMC 29 times and emcee/AIES 23 times prior to the predetermined maximum number of likelihood evaluations (i.e. 5×10^3 in this case). Detailed results about the values of the used metrics are shown in Table 2. The 1-D and 2-D marginal posterior distributions are shown in Figure 17, demonstrating the agreement between the three methods.

4 DISCUSSION

Following the analysis we conducted in Section 3 using the normal distribution there are two important questions that need to be answered about the initialisation of the walkers. First, how many walkers are necessary and, second, how to choose the initial positions of the walkers. Although there are many ways of answering those questions and there is no consistent solution that works for all target distributions, we will try to provide some general rules and heuristics to help ease the task of choosing the number and initial positions of the walkers for most cases.

Let us first discuss the effect of the number of the walkers on the general performance of zeus. Naively, one might expect that the minimum number of walkers should be $D + 1$, where D is the number of dimensions. However, the ensemble splitting technique, which was introduced in Section 3 to render the algorithm parallelisable, requires at least $2 \times D$ walkers in order to produce 2 linearly independent samples. If a smaller number is chosen then the walkers can be trapped in a lower–dimensional hyper–plane of the parameter space, being unable to sample properly and leading to erroneous results. Although there is no upper bound on the number of walkers, we recommend to use between two to four times the number of dimensions. The reason is that increasing the number of dimensions can increase the cost of the burn-in period as we explained in detail in Section 3. Ideally, one wants to use the minimum number (or close to that) of walkers until the burn-in period is over and then increase the number of walkers to rapidly produce a great number of independent samples. It is also worth noting that in cases in which either non-linear correlations or multiple modes are present it is recommended to use more walkers (e.g. 4-8 times the number of parameters for a bimodal target distribution).

As for the initialisation of the walkers, there are many ways to choose their starting positions ranging from prior sampling to more

localised initial positions. Empirical tests indicate that the latter often outperforms the former (i.e. leads to shorter burn-in periods). That is not surprising since the total probability of a prior-sampled initialisation can be very small when the number of parameters is high. In particular we found that initialising the walkers from a tight region in parameter space (i.e. normal distribution with small variance) consistently leads to good performance. For low to moderate dimensional problems initialising the walkers from a tight ball around the *Maximum A Posteriori* (MAP) estimate can substantially reduce the burn-in period (Foreman-Mackey et al. 2013).

Finally, while `emcee`/AIES and `emcee`/DEMC can sample acceptably from most target distributions with $D \lesssim 20$, the efficient scaling of `zeus` with the number of parameters allows us to extend this range and efficiently test more complicated models (Karamanis & Beutler 2020). Like most gradient-free methods, `zeus` will fail to sample efficiently in very high dimensional problems in which $D = O(10^2)$. In such cases, more sophisticated algorithms (e.g. tempering, block updating, Hamiltonian dynamics etc.) need to be used (Robert et al. 2018).

5 CONCLUSIONS

The aim of this project was to develop a tool that could facilitate Bayesian parameter inference in computationally demanding astronomical analyses and tackle the challenges posed by the complexity of the models and data that are often used by astronomers. To this end, we introduced `zeus`, a parallel, general-purpose and gradient-free `Python` implementation of Ensemble Slice Sampling.

After introducing the method in Section 2, we thoroughly demonstrated its performance compared to two popular alternatives (i.e. `emcee` with affine-invariant ensemble sampling and differential evolution Metropolis) using a variety of artificial and realistic target distributions in Section 3. The artificial toy examples helped to shed light on the general behaviour of the samplers in target distributions characterised by linear and non-linear correlations as well as multimodal densities. When compared to `emcee`/AIES and `emcee`/DEMC in the problems of Baryon Acoustic Oscillation parameter inference and exoplanet radial velocity fitting, `zeus` consistently converges faster (i.e. its burn-in is shorter by a factor of at least 3), it is less sensitive to the initialisation of the walkers and generates substantially more independent samples per likelihood evaluation (i.e. approximately $\times 9$ and $\times 29$ speed-up compared to `emcee`/AIES in the cosmological and exoplanet examples, respectively).

We have shown that `zeus` performs similarly or better than existing MCMC methods in a range of problems. We hope that `zeus` will prove useful to the astronomical and cosmological community by complementing existing approaches and facilitating the study of novel models and data over the coming years. `zeus` is publicly available at `https://github.com/minaskar/zeus` with detailed documentation and examples that can be found at `https://zeus-mcmc.readthedocs.io`.

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`zeus` and this work have benefited from a variety of `Python` packages including `numpy` (Van Der Walt et al. 2011), `scipy` (Virtanen et al. 2020), `matplotlib` (Hunter 2007), `seaborn` (Waskom 2021), `getdist` (Lewis 2019), `sklearn` (Pedregosa et al. 2011), `tqdm` (da Costa-Luis 2019), and `mpi4py` (Dalcin et al. 2011).

DATA AVAILABILITY

All data used in this work are publicly available. Power spectrum estimates, covariance matrices and window functions used in the cosmological inference example are available at `http://www.sdss3.org/science/boss_publications.php`. Radial velocity measurements of the K2-24 system used in the exoplanet inference are available as part of the `RadVel` package at `https://github.com/California-Planet-Search/radvel`.

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APPENDIX A: CRONUS

Along with zeus, we introduce cronus, a Python package that wraps zeus as well as other samplers (e.g. emcee). cronus is designed to be used via the terminal through simple, human-readable parameters files. cronus implements an automated suite of convergence diagnostics and all its methods are massively parallel using the Message Passing Interface (MPI). Our aim is to automate parameter inference and facilitate highly reproducible analyses for current and future astronomical surveys such as the Dark Energy Spectroscopic Instrument (DESI) (DESI Collaboration et al. 2016) cronus is available at https://cronus-mcmc.readthedocs.io.

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